

Winston Zhang

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seascorpion.github.io/Portfolio

Final-year Mechanical Engineering Honours student with expertise in mechanical design, CAD, FEA, CFD, and manufacturing. Experienced in leading projects from concept to validation, including 3D CAD modelling, finite element analysis, fluid dynamics simulation, and rapid prototyping. Strong background in cross-functional teamwork, DFM, tolerance analysis, and structured verification and validation.

EXPERIENCE

SYDNEY MOTORSPORT FORMULA STUDENT TEAM

Advisor

February 2025 - Present

Working with the aerodynamics team to validate the full car aerodynamic package using the University of Sydney's wind tunnel as part of graduate thesis project. Provided technical guidance on CFD validation and composite manufacturing, utilising in-house knowledge and development over the 4 years with the team.

Aerodynamics Section Lead

December 2023 - February 2025

Led a team of 10 student engineers for the in-house development and manufacturing of a full aerodynamics package using SOLIDWORKS, ANSYS Fluent, laser cutting, and 3D printing. We delivered a structurally robust aerodynamic package with improved structural rigidity over previous years while maintaining a strict 3 kg weight target. This required the design of a rib and spar system for the front wing to handle both expected and unexpected loads during testing and competition. Coordinated the large full car assemblies (>1000 parts) to ensure team design cohesion, enforced compliance with FSAE rules, and managed my personal and teams timelines and task delegation to ensure the car was delivered on time.

Aerodynamics Engineer

July 2022 - December 2023

Designed and validated the front wing and under-tray of the SM23 vehicle using CFD within ANSYS Fluent. Conducted an iterative design processes using a large computing cluster to optimize aerodynamic performance. Participated in mold design and manufacturing, and performed carbon fiber layups using pre-impregnated and conventional wet-layup techniques to produce high-performance composite component.

DOVETAIL ELECTRIC AVIATION

Computational Fluid Dynamics Engineer (Intern)

September 2023 - January 2024

Reverse-engineered a propeller nacelle assembly by generating parametric CAD models from 3D scans in SOLIDWORKS. Validated scanned propeller geometry in ANSYS Fluent, allowing for the optimization of cooling duct geometries to be developed, minimizing drag and improve airflow duct efficiency. Created efficient lightweight CFD models to reduce simulation time and resource consumption, without impact to data resolution.

NANOSONICS

Warehouse Operator

Aug 2025 - Present

Supported production and dispatch operations by accurately picking and packing materials, performing quality inspections of incoming and outgoing goods, and managing the shipping and receiving of large, delicate scientific instruments. Maintained reliable warehouse stock records through internal systems to ensure traceability, and applied the ability to interpret technical drawings for correct part verification, contributing to smooth manufacturing workflows and strict compliance with handling standards.

TECHNICAL SKILLS

Software

- Created mechanical designs with DFM principles in **SOLIDWORKS** and **Autodesk Inventor**
- Conducted various numerical simulations within **SOLIDWORKS** and the **ANSYS suite** such as **FEA**, **topology studies**, **CFD** and **thermal analysis**
- Developed **MATLAB** scripts for numerical simulations as well as various data-parsing programs.
- Proficient in **Python** through various calculation and simulation scripts.

Manufacturing

- Carbon fiber composite parts with **pre-impregnated** and conventional **wet-layup** techniques
- Create composite molds with contemporary techniques such as **hot wire cutting**, **laser cutting**, **3D printing**, along with conventional techniques such as **hand-sculpting** foam molds
- 3D Printing** and **laser cutting** for rapid design and manufacturing
- Extensive experience in common workshop tooling, and machinery

TECHNICAL PROJECTS

- **Homelab Server** - Functions as both a local media server and a self-hosted cloud storage solution, enabling high-speed remote access and data redundancy.
- **Robotic Arm** - A 6-axis robotic arm powered by NEMA stepper motors, with motion transmitted via belt drives to reduce backlash. The accuracy was enhanced through a custom-developed zero-backlash cycloidal gearbox tailored for precise torque and speed control.
- **Custom PC Peripherals** - Engineered a 32-gram fingertip grip mouse optimized for competitive FPS gaming, and assembled a Lily58 split ortholinear keyboard with a custom-designed case for improved ergonomics and portability.
- **UNIX** - Daily-operated dual-boot Windows/Linux setups, leveraging distros such as Arch and Ubuntu derivatives for development flexibility, customization, and deep system-level understanding.

EDUCATION

THE UNIVERSITY OF SYDNEY

Bachelor of Mechanical Engineering Honours (Fluid Mechanics)

Expected Graduation: 2025

2021 - Current